

Web Technology



React – Hooks Lifecycle and useEffect

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Front End Development Tasks

App state	Data definition, organization, and storage
User actions	Event handlers respond to user actions
Templates	Design and render HTML templates
Routing	Resolve URLs
Data fetching	Interact with server(s) through APIs and AJAX

User Actions

1. UI Reacts to User Events

- Clicks, Typing, or Form Submissions using Event Handlers
- `onClick`, `onChange`

2. Front End Task

- Handling button clicks to submit data or toggle views
- Capturing and updating form input values
- Navigating between pages or sections based on user interaction

React State and Lifecycle

1. **State**
2. **Lifecycle**

React Hooks

1. **Hooks – Enables**
 - **States**
 - **Lifecycle**
2. **Eliminates need for class component**
3. **Types of Hooks**
 - **useState**
 - **UseEffect, useContext**
 - **useReducer**
 - **useMemo**
 - **useRef**
 - **useCallback**

React Hooks

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 - Different Stages of a Component

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 1. Mounting
 2. Updating
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 1. Mounting – Appears on Screen
 2. Updating – Re-renders due to prop or State Change
 3. Unmounting – When Removed

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 - Operations Outside Component – Side Effects

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 - Operations Outside Component – Side Effects
 1. Eg: Fetching Data from API, Logging Events, Direct DOM Manipulation
 2. Achieved using useEffect

React Hooks Lifecycle vs States

Aspect	Lifecycle	State
What it represents	Stages/events in component's existence (mount, update, unmount)	Data or variables tracking component's dynamic values
Purpose	To run side-effects (e.g., fetching data, subscriptions, timers) at specific times	To hold and manage data that affects rendering and UI behavior
Managed by	<code>useEffect</code> Hook (or other lifecycle-related hooks)	<code>useState</code> Hook
Changes cause	Effects like API calls, cleanup, updates outside rendering	Re-render of component with new state values
Example usage	Fetch data on mount, set up event listener, clean up on unmount	Track form input value, toggle a modal, count clicks

React Hooks - useEffect

1. **State Update Trigger Lifecycle Effects**
2. **Lifecycle Effects Trigger State Update**

React Hooks - useEffect

1. **Observer**
2. **Runs based on Side effects Change**

React Hooks - useEffect

1. **Observer**
2. **Runs based on Value Change**
3. **Syntax: `useEffect(<function>, <dependency>)`**
4. **Empty Dependency `[]` - Triggers Only Once on Re-render**

```
useEffect(() => {  
  console.log(`myValue was changed! New value: ${myValue}`);  
}, [myValue]);
```



A list of values such that changes
should trigger this code to run



The code to run when
values change

React Hooks – Online Sources

1. `import useEffect from 'react'`
2. **Identify and Set Dependency**
3. `useEffect` logic
4. **Test `useEffect`**

`useEffect` → "What should my component do after rendering or when something changes?"

```
useEffect(() => {  
  document.title = `Count: ${count}`;  
}, [count]);
```

Here:

- `count` changes → component re-renders
- After rendering → `useEffect` runs
- It changes the **browser tab title**

React Hooks – Online Sources

1. React useEffect Hooks

- useEffect – React

References

1. **Robin Wieruch, “The Road to React: The ReactJS with Hooks”, Lean Publishing, 2023 Edition.**
2. **Slides – Aaron Smith - University of North Carolina, Chapel Hill**

THANK YOU

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